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Article

The risk and benefit profiles of US-eligible lung cancer screening attendees vs nonattendees

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Abstract

Background: The US Preventive Services Task Force (USPSTF) recommend lung cancer screening for individuals aged 50-80 years with at least 20 pack-years and no more than 15 quit-years, but uptake is low. The risk and benefit profiles of screening attendees are unknown; consequently, the impact and lost opportunity of ongoing lung cancer screening in the United States remains unclear.

Methods: We estimated lung cancer death risk (using the Lung Cancer Death Risk Assessment Tool) and life gained from screening (using the Life Years Gained From Screening-Computed Tomography model) for individuals aged 50-79 years who ever-smoked in the US representative 2022 Behavioral Risk Factor Surveillance System. We compared lung cancer death risk and life gained among USPSTF-eligible individuals by screening status (self-reported screened vs not screened in past year) and estimated the number of lung cancer deaths averted and life-years gained under current screening levels and if everyone eligible was screened.

Results: USPSTF eligibility was 33.7% (95% confidence interval [CI] = 33.1% to 34.4%), of whom 17.9% (95% CI = 17.0% to 18.8%) self-reported screening. Screening uptake increased with increasing lung cancer death risk quintile (Q1 = 5.2%, 95% CI = 3.0% to 8.8%; Q5 = 21.8%, 95% CI = 20.3% to 23.3%) and life-gain from screening quintile (Q1 = 6.2%, 95% CI = 3.8% to 9.9%; Q5 = 20.8%, 95% CI = 19.5% to 22.2%). Screened individuals had higher lung cancer death risk (risk ratio [RR] = 1.35, 95% CI = 1.26 to 1.46) and life-years gained (RR = 1.19, 95% CI = 1.12 to 1.25) than unscreened individuals. Currently, screening averts 19 306 lung cancer deaths and gains 237 564 life-years; screening everyone eligible would additionally avert 56 956 lung cancer deaths and gain 751 850 life-years. Two-thirds of USPSTF lung-eligible women were up to date with breast cancer screening, but only 17.3% attended lung screening in the past year.

Conclusions: Eligible screening attendees had higher lung cancer death risk and benefit from screening. Higher rates of screening could substantially increase the number of lung cancer deaths prevented.

Lung cancer is the leading cause of cancer death in the United States for men and women, with an estimated 127 070 deaths from lung cancer occurring in the United States in 2023 (1). Lung cancer screening has been recommended by the US Preventive Services Task Force (USPSTF) since 2013, based on evidence from the National Lung Screening Trial (NLST), which showed that 3 annual low-dose computed tomography (CT) screens reduced 5-year lung cancer mortality by 20% (2). In the 2013 guidelines, individuals aged 55-80 years with at least 30 pack-years and who either currently smoke or quit within the past 15 years were eligible. In 2021, USPSTF updated their guidelines, expanding eligibility to individuals aged 50-80 years with 20 or more pack-years who either currently smoke or quit within the past 15 years. However, recent publications have reported low and variable uptake rates of lung cancer screening in the United States among those eligible for screening (3-7).

Many factors influence the probability of an eligible person attending lung cancer screening, including race and ethnicity (7,8), health status and number of comorbidities (8,9) and insurance status (4,5). However, it is not known whether those attending screening are the highest risk or highest benefit individuals and therefore what proportion of preventable lung cancer deaths are being prevented by screening and what proportion of gainable life-years are being gained. We therefore investigated the risk and benefit profiles of lung cancer screening attendees and quantified the lost opportunity resulting from the low uptake of screening.

Methods

Data

The Behavioral Risk Factor Surveillance System (BRFSS) is an annual nationwide state-based telephone survey that collects self-reported sociodemographic and health-related information, providing nationally representative estimates for the US population (10). All BRFSS participants provided informed consent. In 2022, the median survey response rate for all states; territories; and Washington, DC, was 45.1% (range = 22.8%-66.8%) (11).

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Although the lung cancer screening module was first introduced in BRFSS in 2017, it was not included as part of the core survey, asked to all states, until 2022. Thus, we used the 2022 survey to investigate lung cancer screening uptake in the United States among individuals aged 50-79 years who self-reported having ever-smoked 100 cigarettes, with no prior history of lung cancer. Individuals who reported currently or formerly smoking were asked whether they had had a CT or computed axial tomography scan of the chest area mainly to check or screen for lung cancer and, if so, time since their most recent scan.

Exposures of interest

We used the Lung Cancer Risk Assessment Tool, the Lung Cancer Death Risk Assessment Tool, and the Life Years Gained From Screening–CT (LYFS-CT) models to obtain estimates for 5-year risk of lung cancer incidence, 5-year risk of lung cancer mortality, and individualized life gained from an NLST-like CT screening program, respectively (12,13). Each of these models has been validated in the US population (12,13). We assumed that attending 3 annual lung cancer screens resulted in a 20.4% reduction in lung cancer mortality for 5 years, as observed in the NLST (2). All models were implemented using the *lcmodels* package in R (14). For details on the variables used in the models, see the Supplementary Material (available online). All analyses accounted for BRFSS's complex survey sample design, in accordance with BRFSS guidelines (10).

Statistical analysis

We used multiple imputation to account for item-level missing data (0%-12.5%) for all individuals aged 50-79 years without a prior history of lung cancer who had ever-smoked; see Supplementary Table 1 (available online) for levels of missingness and the Supplementary Material (available online) for imputation details.

We carried out descriptive analyses of the population of individuals aged 50-79 years who ever-smoked, reporting the number of individuals and the weighted proportion of individuals with each characteristic. For individuals eligible for lung cancer screening under USPSTF-2021, we reported the median and interquartile range (IQR) of 5-year lung cancer risk (from Lung Cancer Risk Assessment Tool), 5-year lung cancer death risk (from Lung Cancer Death Risk Assessment Tool), and life-years gained (benefit) from attending screening (from LYFS-CT) for individuals who did and did not attend lung cancer screening in the past year. These were reported overall and stratified by a range of characteristics. Additionally, we calculated the ratio of the medians for lung cancer death risk and benefit from attending screening for individuals who attended screening in the past year compared with those who did not (see Supplementary Material, available online for more details). All tests of statistical significance were 2-sided, at a level of 95% significance.

The same analyses were carried out for a subset of characteristics for individuals who were ineligible for screening under USPSTF-2021. All analyses were also repeated using an outcome of ever attending lung cancer screening. We carried out 2 sensitivity analyses. First, we used a complete case analysis (removing adults with missing values for any smoking variables, lung cancer screening, or sex), using multiple imputation for all remaining missing data using the default method provided by the *lcmodels* package (14). In the second, we removed adults with a prior history of any cancer, excluding skin cancer.

Finally, we estimated the lost opportunity because of limited uptake of lung cancer screening. We evaluated 5 screening uptake scenarios within our sample of individuals aged 50-79 years who had ever-smoked, without a previous history of lung cancer: (1) those who reported ever being screened in the past year, (2) those eligible under USPSTF-2013 who did not report screening, (3) individuals eligible under USPSTF-2021 but not USPSTF-2013 who did not report screening, (4) individuals eligible under the American College of Chest Physicians (ACCP) guidelines (≥16.2 gainable life-days) (15) who were ineligible under USPSTF-2021 and who did not report screening, and (5) the remaining individuals. Under each uptake scenario, we quantified the proportion of additional individuals screened and the modeled proportion of additional preventable lung cancer deaths averted over 5 years and years of life gained over the full life course. We additionally provide these estimates under assumptions of 25%, 50%, and 75% screening uptake in each of these categories, assuming those who are screened are a random selection of the individuals in the category.

All analyses were conducted in R v4.2.0 and v4.2.2.

Results

Among BRFSS 2022 respondents, 97 856 participants were aged 50-79 years without a history of lung cancer and reported having smoked at least 100 cigarettes, representing 44 000 060 individuals in the United States; their characteristics are given in Table 1. Of note, 16.1% were aged 50-54 years and could only have become eligible for screening in 2021 (or 2022 for those aged 50 years in 2022) under the 2021 guidelines, because the 2013 guidelines restricted screening to ages 55-80 years. The majority did not currently smoke at the time they participated in BRFSS (69.8%), with half of individuals aged 50-79 years who ever-smoked having smoked less than 20 pack-years, the USPSTF-2021 threshold.

Screening uptake in the past year

During 2022, 33.7% (95% confidence interval [CI] = 33.1% to 34.4%) of individuals aged 50-79 years who ever-smoked were USPSTF-2021 eligible, including 20.8% (95% CI = 20.3% to 21.4%) who were USPSTF-2013 eligible. Among those eligible under USPSTF-2021 guidelines, 17.9% (95% CI = 17.0% to 18.8%) self-reported undergoing a low-dose CT to check or screen for lung cancer (from here on referred to as screening) in the past year.

The proportion screened was similar for men (18.4%, 95% CI = 17.1% to 19.7%) and women (17.3%, 95% CI = 16.1% to.18.6%) and increased with increasing age, from 6.9% (95% CI = 5.5% to 8.8%) for individuals aged 50-54 years to 26.7% (95% CI = 23.0% to 30.9%) for individuals aged 75-79 years (Figure 1, A). Screening uptake was similar across races and ethnicities (African American: 20.7%, 95% CI = 16.9% to 25.2%; Hispanic American: 16.4%, 95% CI = 12.7% to 21.1%; non-Hispanic White: 17.9%, 95% CI = 17.0% to 18.8%; Other [ie, not non-Hispanic White, non-Hispanic Black, or Hispanic]: 15.7%, 95% CI = 11.3% to 21.4%). Screening uptake was highest in the Northeast (21.6%, 95% CI = 19.2% to 24.2%) and lowest in the territories (12.6%, 95% CI = 7.7% to 20.0%) (Supplementary Table 2, available online). Screening uptake in the past year was only 4.1% (95% CI = 2.6% to 6.4%) among individuals without insurance and was higher among individuals with government insurance (20.9%, 95% CI = 19.8% to 22.1%) than private insurance (14.2%, 95% CI = 12.7% to 15.8%). Among individuals without a medical checkup in the past year, only 5.2% (95% CI = 4.0% to 6.8%) had attended screening.

The proportion who attended screening increased with both increasing lung cancer death risk and life-years gained from

Table 1. Demographic characteristics of Behavioral Risk Factor Surveillance System 2022 individuals aged 50-79 years who ever-smoked, with no prior diagnosis of lung cancer

	Not screened in the past year		Screened in the past year		Missing screening status in the past year		Total	
Characteristics	Unweighted, No.	Weighted, %	Unweighted, No.	Weighted, %	Unweighted, No.	Weighted, %	Unweighted, No.	Weighted %
Total	78 977	80.2	8728	9.1%	10 151	10.8	97 856	100
Sex								
Men	39 312	52.3	4578	54.6	5072	51.0	48 962	52.4
Women	39 665	47.7	4150	45.4	5079	49.0	48 894	47.6
Age, y								
50-54	10 748	17.6	444	7.0	977	12.8	12 169	16.1
55-59	11 899	17.6	1012	13.9	1212	13.3	14 123	16.8
60-64	14 902	21.4	1811	22.8	1774	20.1	18 487	21.3
65-69	15 743	17.2	2175	21.4	1999	18.4	19 917	17.7
70-74	14 377	15.3	1975	19.7	2154	19.4	18 506	16.2
75-79	11 308	10.8	1311	15.2	2035	16.0	14 654	11.8
Self-reported race and ethnicity								
African American	4935	9.4	684	11.4	914	12.9	6533	10.0
Hispanic	4243	10.3	328	8.1	499	9.2	5070	10.0
Non-Hispanic White	65 331	73.0	7271	74.1	8170	70.3	80 772	72.8
Other[a]	4468	7.3	445	6.4	568	7.5	5481	7.3
Region								
Midwest	21 221	22.3	2533	24.5	2704	21.2	26 458	22.4
South	22 170	37.5	2664	37.2	3302	39.9	28 136	37.7
Northeast	15 345	17.7	1883	21.3	1932	17.4	19 160	18.0
West	19 086	21.6	1525	16.3	2099	20.8	22 710	21.0
Territory	1155	0.9	123	0.6	114	0.7	1392	0.9
Smoking status								
Former	57 931	71.2	4831	57.0	7076	70.1	69 838	69.8
Current	20 923	28.8	3873	43.0	2934	29.9	27 730	30.2
Missing	123		24		141		288	
Pack-years								
<10	22 869	33.2	934	12.9	1894	27.7	25 697	30.8
10-19	15 260	21.5	1166	15.0	1449	21.1	17 875	20.8
20-29	11 469	15.8	1562	18.0	1222	16.6	14 253	16.1
30-49	13 204	18.8	2343	27.6	1452	19.9	16 999	19.7
≥50	7535	10.7	2145	26.5	1151	14.8	10 831	12.6
Missing	8640		578		2983		12 201	
Healthcare coverage								
Private insurance	26 216	38.3	1910	27.8	2418	28.7	30 544	36.3
Government insurance	47 786	56.6	6495	71.0	7002	67.4	61 283	59.0
No insurance	2901	5.2	75	1.2	288	3.9	3264	4.7
Missing	2074		248		443		2765	
Had a checkup in the past year								
No	12 186	16.5	380	4.8	1218	13.0	13 784	15.1
Yes	66 195	83.5	8297	95.2	8791	87.0	83 283	84.9
Missing	596		51		142		789	
Annual household income, US$								
0-19 999	8978	14.4	1452	19.9	1488	20.5	11 918	15.5
20 000-49 999	22 879	32.7	3065	40.9	2906	35.9	28 850	33.7
50 000-99 999	20 600	29.3	1965	24.5	2072	26.5	24 637	28.6
≥100 000	14 488	23.7	935	14.7	1207	17.1	16 630	22.2
Missing	12 032		1311		2478		15 821	

[a] All races and ethnicities other than African American, Hispanic, and non-Hispanic White.

screening quintile, from 5.2% (95% CI = 3.0% to 8.8%) to 21.8% (95% CI = 20.3% to 23.3%) with increasing lung cancer death risk and from 6.2% (95% CI = 3.8% to 9.9%) to 20.8% (95% CI = 19.5% to 22.2%) with increasing benefit from attending screening (Supplementary Table 2, available online). Although screening guidelines advise not to screen individuals with a limited life expectancy (16,17), a quarter of individuals with a life-expectancy no more than 5 years attended screening in the past year (24.9%, 95% CI = 15.6% to 37.3%), though only 1.0% of the individuals in this study had a life expectancy no more than 5 years.

The median 5-year lung cancer death risk among USPSTF-2021–eligible individuals who had attended screening in the past year was 2.00% (95% CI = 1.03% to 3.93%) compared with 1.48% (95% CI = 0.71% to 2.99%) (Supplementary Table 2, available online) among USPSTF-2021–eligible individuals not screened in the past year (risk ratio [RR] = 1.35, 95% CI = 1.26 to 1.46; $P < .001$), indicating that among those who were USPSTF-2021–eligible,

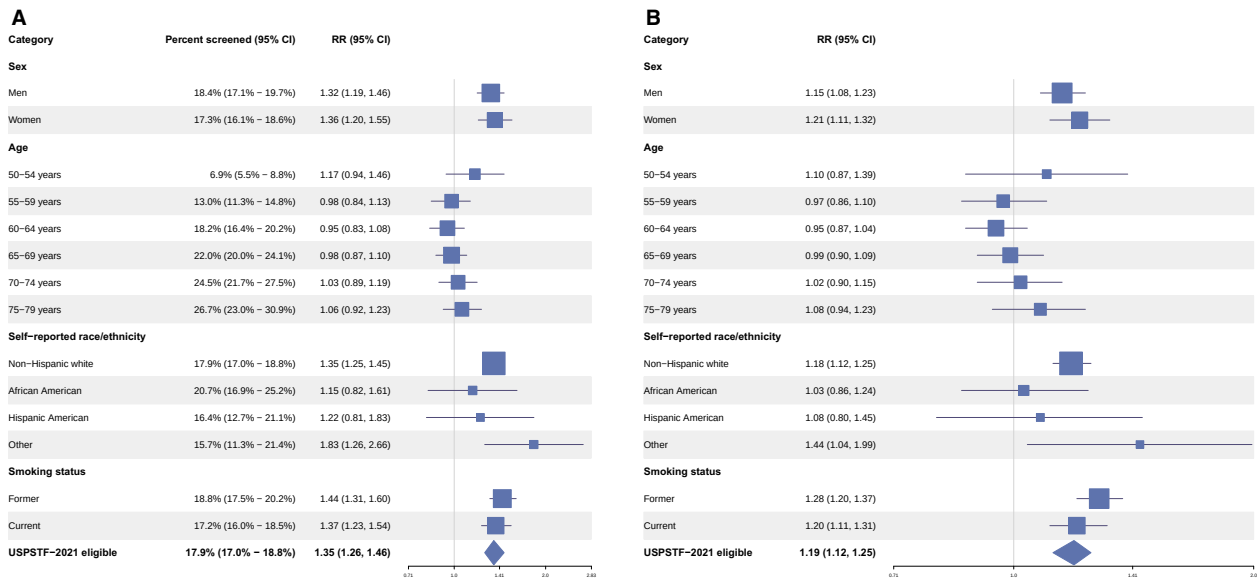


Figure 1. A forest plot showing the percentage of US Preventive Services Task Force-2021–eligible individuals who attended screening by sex, age, self-reported race and ethnicity, and screening status, as well as (**A**) the relative median lung cancer death risk and (**B**) the relative median life gained from attending screening, among screening attenders compared with nonattenders. CI = confidence interval; RR = risk ratio; USPSTF = US Preventive Services Task Force.

screened individuals were higher risk than those who were not screened (Figure 1, A; Supplementary Table 3, available online). Among individuals who identified as "Other" race and ethnicity, screened individuals had 1.83 (95% CI = 1.26 to 2.66) times the median risk of individuals who were not screened in the past year. USPSTF-2021–eligible individuals in the Western United States who were screened also had a much higher median risk than unscreened individuals (RR = 1.70, 95% CI = 1.29 to 2.23), as did individuals with an annual household income of $100 000 or more (RR = 1.83, 95% CI = 1.41 to 2.36). The ratio was slightly below 1 (indicating a lower median risk for individuals who were screened compared with those who were not screened) for individuals in the lowest lung cancer death risk quintile, those with 5 or more comorbidities, and individuals aged 55-69 years, though none of these were statistically significant.

Similarly, the median predicted days of life gained among USPSTF-2021–eligible individuals screened in the past year (22.09, IQR = 13.92-34.66) was higher than the predicted anticipated benefit among individuals who weren't screened (18.65, IQR = 10.99-30.33; RR = 1.19, 95% CI = 1.12 to 1.25). The median days of life gained for USPSTF-2021–eligible individuals who were and were not screened by characteristic are shown in Supplementary Table 2 (available online), with the risk ratios shown in Figure 1, B, and Supplementary Table 3 (available online).

Among individuals aged 50-79 years who ever-smoked but were ineligible under USPSTF-2021 guidelines, 6.4% (95% CI = 5.9% to 6.9%) reported screening in the past year, corresponding to 41% of screens among individuals who ever-smoked. The percentage screened increased with increasing age, lung cancer death risk quintile, and benefit from screening quintile (Supplementary Table 4, available online). Ineligible individuals who were screened had higher median lung cancer death risk (RR = 1.78, 95% CI = 1.60 to 1.97) and benefit from screening (RR = 1.52, 95% CI = 1.37 to 1.68) compared with ineligible people who were not screened, though their median modeled absolute lung cancer death risk (0.61%, IQR = 0.29%-1.30%) and days of life gained from screening (7.64, IQR = 4.39-13.81) were modest (Supplementary Table 5, available online). Of note, 6.4% of individuals who had never-smoked reported ever being screened, including 2.7% screened in the past year.

We additionally compared breast cancer screening uptake among women eligible for lung cancer screening under USPSTF-2021; 94.7% had ever received mammography, including 67.8% within the past 2 years.

Sensitivity analyses

Overall, results were robust in sensitivity analyses. In the complete case analysis (Supplementary Tables 6 and 7, available online), 18.5% (95% CI = 17.6% to 19.5%) of eligible adults attended screening in the past year compared with 17.9% in our main analysis. Risk profile comparisons between individuals who did and did not screen were similar. When restricted to individuals with no prior cancer (Supplementary Tables 8 and 9, available online), a slightly lower proportion of USPSTF-2021–eligible individuals had been screened in the past year (15.5%, 95% CI = 14.5% to 16.4%).

Among individuals eligible under USPSTF-2021, 31.0% (95% CI = 29.9% to 32.2%) had ever been screened. More detailed results with an outcome of lifetime screening uptake are provided in the Supplementary Material (Supplementary Tables 10-17, available online).

Lost opportunity

In 2022, 10.3% of individuals aged 50-79 years who ever-smoked reported screening in the past year (n = 4 517 396), which averted an estimated 17.6% of preventable lung cancer deaths and gained 15.7% of gainable years of life (Figure 2, Table 2). If everyone eligible under USPSTF-2013 guidelines had attended screening, 16.5% more individuals would be screened, averting 42.0% more of the preventable lung cancer deaths and gaining 38.1% more gainable years of life. Additionally, screening the 11.2% of individuals who

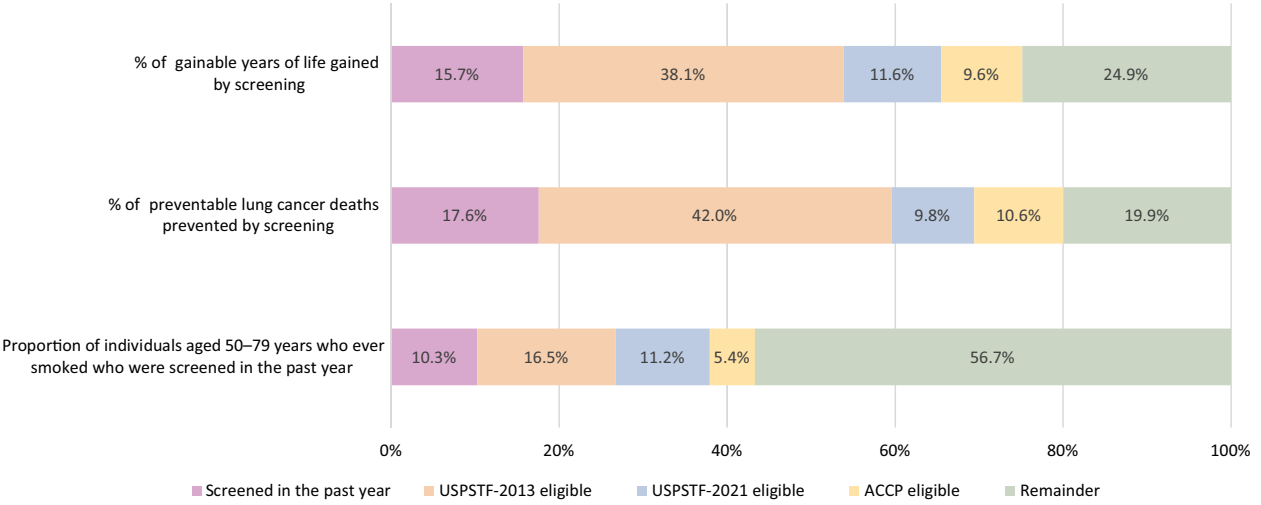


Figure 2. The proportion of individuals, lung cancer deaths prevented, and life-years gained from attending screening by screening (1) everyone screened in the past year, (2) additionally screening everyone eligible under US Preventive Services Task Force (USPSTF)–2013, (3) additionally screening everyone eligible under USPSTF-2021, (4) additionally screening everyone eligible under the American College of Chest Physicians (≥16.2 days of life gained from attending screening), and (5) the remaining individuals aged 50-79 years who ever-smoked. ACCP = American College of Chest Physicians.

Table 2. The number of people who would be screened, number of lung cancer deaths prevented, and years of life gained under five screening uptake scenarios

If the following individuals were screened	No. of people who would be screened	% of individuals aged 50-80 years who ever-smoked who would be screened	Lung cancer deaths prevented by screening	% of preventable lung cancer deaths prevented by screening	% of all lung cancer deaths prevented by screening	Years of life gained from screening	% of years of life gained from screening
Those who reported screening in the past year	4 517 396	10.3	19 306	17.6	3.6	237 564	15.7
+[a] those eligible under US Preventive Services Task Force 2013	11 771 215	26.8	65 452	59.6	12.2	813 643	53.9
+ those eligible under US Preventive Services Task Force 2021	16 693 109	37.9	76 262	69.4	14.2	989 413	65.5
+ those eligible under American College of Chest Physicians[b]	19 051 544	43.3	87 948	80.1	16.3	1 134 864	75.1
+ everyone else aged 50-79 years who ever smoked	44 000 060	100.0	109 850	100.0	20.4	1 510 543	100.0

[a] "+" indicates that the row includes all those in the previous row(s) as well as those referenced in the current row.
[b] At least 16.2 days of life gained according to the Life Years Gained from Screening–Computed Tomography model.

are eligible under USPSTF-2021 guidelines would avert an additional 9.8% of lung cancer deaths and gain 11.6% of gainable years of life. If everyone who is estimated to gain at least 16.2 days of life under LYFS-CT were screened, per ACCP guidelines, an additional 5.4% of individuals aged 55-79 years who ever-smoked would be screened beyond those eligible under USPSTF-2021 guidelines, averting an additional 10.6% of preventable lung cancer deaths over 5 years and gaining an additional 9.6% of gainable years of life. Only 19.9% of preventable lung cancer deaths and 24.9% of gainable years of life occur among the 56.7% of individuals aged 55-79 years who ever-smoked who would not be eligible under any of these criteria.

We note that only 20.4% of lung cancer deaths are preventable by screening, therefore current screening is preventing only 3.6% of lung cancer deaths; screening everyone eligible under USPSTF-2021 would prevent 14.2% of lung cancer deaths.

Results assuming 25%, 50%, and 75% uptake among each of the additional groups of currently unscreened individuals are

shown in Supplementary Table 18 (available online). If 50% of people in each group were screened in addition to those screened in the past year, 55.1% of individuals aged 50-80 years who ever-smoked would be screened, preventing an estimated 58.8% of preventable lung cancer deaths over 5 years and gaining 57.9% of gainable life-years.

Results by US state

The proportion of USPSTF-2021–eligible individuals who were screened in the past year varied by state (Figure 3, A; Supplementary Table 19, available online), from 10.3% (95% CI = 7.2% to 14.5%) in Wyoming to 27.8% (95% CI = 22.3% to 34.1%) in Rhode Island. Certain states screened higher risk and higher benefit USPSTF-2021–eligible individuals (ratio of median lung cancer death risk among those screened compared with unscreened [Figure 3, B; Supplementary Table 20, available online]: Hawaii: 2.43, 95% CI = 1.22 to 4.82; Nevada: 2.25, 95% CI = 0.98 to 5.16; Wyoming: 2.09, 95% CI = 1.35 to 3.24; ratio of median life-years gained from attending screening among those screened compared with not screened [Figure 3, C; Supplementary Table 20, available online]: Nevada: 1.60, 95% CI = 1.02 to 2.52; South Dakota: 1.58, 95% CI = 0.68 to 3.67; Hawaii: 1.58, 95% CI = 1.08 to 2.30).

Discussion

Using a nationally representative, cross-sectional survey, we report that screening uptake remains low, with 17.9% of eligible individuals self-reporting a low-dose CT to check for lung cancer in the previous year, and only 19.9% of preventable lung cancer deaths are prevented by screening these individuals, showing the lost opportunity to prevent death from the most common cause of cancer death in the United States. We note that among eligible individuals, those who were screened had 35% higher risk of lung cancer death and 19% higher benefit in terms of estimated days of life gained compared to unscreened individuals. Although screening uptake remains low compared with other cancer screening, it is encouraging that the individuals who are being screened have above-average risk and benefit; among individuals aged 50-79 years who ever-smoked who were ineligible under USPSTF-2021, those with a low-dose CT in the past year had 78% higher risk and 52% higher benefit than ineligible individuals who were not screened.

It is encouraging that screening uptake is higher among higher risk and higher benefit individuals, however, 79% of the highest benefit quintile of eligible individuals were not screened in the past year. Identifying these individuals and encouraging them to be screened would have the largest impact on the population-level benefit of screening. We note that more than 90% of individuals aged 50-79 years who ever-smoked reported having health insurance, and 85% had a check-up within the last year, implying missed opportunities for a doctor to discuss lung cancer screening, if appropriate. We note, though, that as of October 2023, lung cancer screening is not yet covered by the Medicaid fee-for-service program in Alabama (18). Among women eligible for lung cancer screening under USPSTF-2021 in BRFSS 2022, 94.7% had ever received mammography, including 67.8% within the past 2 years, showing that this population is not opposed to cancer screening overall. One potential way to improve uptake among women attending mammography would be to assess their lung cancer screening eligibility at their mammography appointment; eligible women could undergo shared decision making, and suitable women would be referred for a lung cancer screen. In BRFSS, only 22% of women eligible for lung cancer screening who were up to date with their breast cancer screening reported attending lung cancer screening in the past year.

If, in addition to everyone who currently reported screening in the past year, everyone eligible under USPSTF-2021 guidelines was screened, only 38% of individuals aged 50-80 years who ever-smoked would have been screened, but 69% of preventable lung cancer deaths would be prevented and 66% of gainable life-years would be gained. If even 50% of USPSTF-2021–eligible individuals were additionally screened, corresponding to 14% more of the 44 million individuals aged 50-80 years who ever-smoked, 44% of preventable lung cancer deaths would be prevented, 41% of gainable life-years gained, showing the incremental benefits of bringing more people into screening.

Individuals at low risk and/or low benefit may experience a net harm from screening; for example, individuals in the lowest lung cancer death risk-quintile in the NLST experienced no reduction in lung cancer mortality (19), though they remain susceptible to the harms of screening (eg, anxiety and further testing resulting from false-positives). Additionally, modeling has estimated that individuals with comorbidities receive less benefit from screening than individuals without comorbidities because of increased competing risks of death and complications from screening (20). Shared decision making is currently recommended before undergoing lung cancer screening, where clinicians can help eligible individuals decide whether they should undergo screening; we therefore note that screening all eligible individuals would not be appropriate.

Recent screening uptake estimates have been variable. The American Lung Association's *2022 State of Lung Cancer* report estimated that only 5.8% of individuals eligible under USPSTF-2013 guidelines were screened in 2021, a much lower estimate than those from self-report sources such as BRFSS. The estimate of 5.8% is from the American College of Radiology's Lung Cancer Screening Registry State Level Comparison for 2021 (3); the report notes that they believe this registry represents most lung cancer screenings, that it was the only approved registry in 2021, and that screening facilities were required to submit data on all lung cancer screenings to meet Medicare eligibility requirements. However, they also note that large, regional managed care providers may not have shared data, which includes the Veterans Affairs, implying that 5.8% is an underestimate. Results from the nationally representative National Health Interview Surveys (NHISs) in 2010 and 2015 reported 3.3% and 3.9%, respectively, of individuals eligible for screening under USPSTF-2013 having low-dose CT screening in the past 12 months (21). In BRFSS 2017-2020, when 8-21 states included the lung cancer screening module, uptake among those eligible under USPSTF-2013 was 12.8%-17.7% (4-6), and in BRFSS 2021 (4 states), uptake was 21.2% of those eligible under USPSTF-2021 (7). Previous research has compared self-reported breast, cervical, and colorectal cancer screening rates reported in NHIS and BRFSS (22), with rates 2%-14% higher (relatively) in BRFSS.

Among USPSTF-2021–eligible individuals, those with no comorbidities had the highest uptake rate. Among individuals with comorbidities, there was an increasing trend with increasing numbers of comorbidities, though the severity of the comorbidities is unknown. Braithwaite and Gould (23) used data from BRFSS 2019 (which included data from 20 states which asked the lung cancer screening module) to evaluate whether screening was reaching the individuals who were most likely to benefit. They used an increasing number of comorbid conditions as a surrogate for decreasing benefit and observed that comorbid

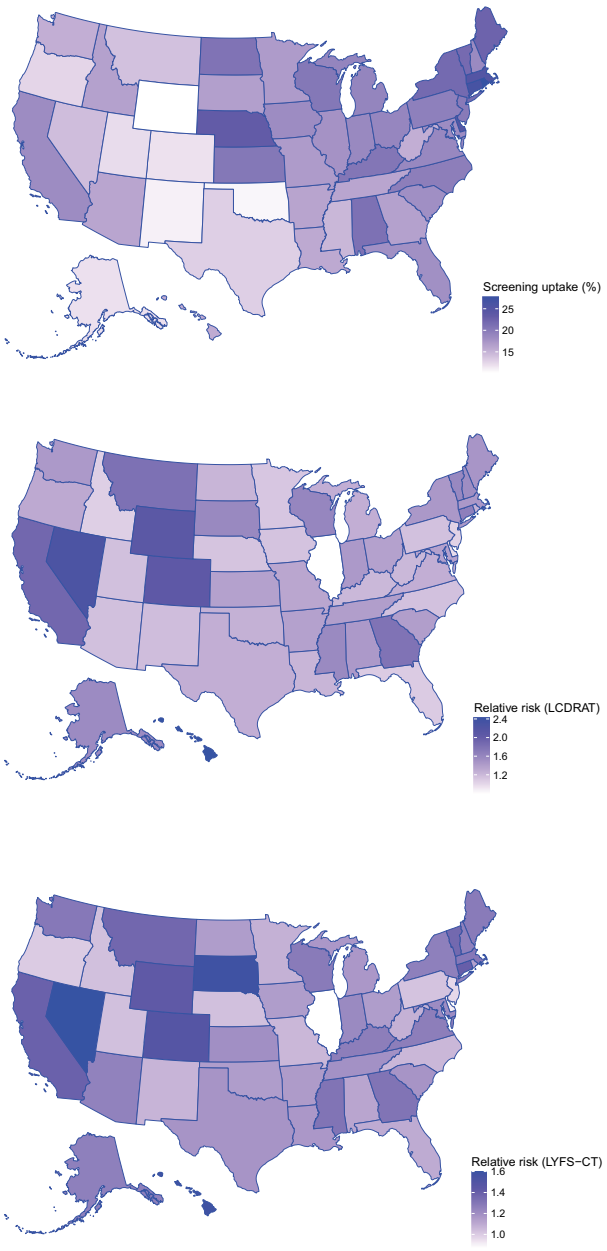


Figure 3. Results by US state: (**A)** lung cancer screening uptake, (**B)** the relative median lung cancer death risk among US Preventive Services Task Force (USPSTF)-2021–eligible individuals who attended screening compared with those who did not, and (**C)** the relative median benefit from attending lung cancer screening among USPSTF-2021–eligible individuals who attended screening compared with those who did not. LCDRAT = Lung Cancer Death Risk Assessment Tool; LYFS-CT = Life Years Gained From Screening–Computed Tomography.

conditions were associated with a higher likelihood of attending screening; similar results were shown by Advani et al. (9). Similarly, Maki et al. (7) and Rustagi et al. (8) showed that lung cancer screening uptake was higher among individuals self-reporting poor health. However, we note that the median benefit among USPSTF-2021–eligible individuals with 5 or more comorbidities who attended screening in the past year was 16.1 days, only just below the ACCP threshold for screening eligibility of 16.2 days. Our study is more optimistic about the life-years gained from current levels of screening, with an estimate approximately 3 times higher than that by Philipson et al. (24), though they used data from 2017 to inform eligibility (ie, prior to the USPSTF-2021 guidelines, which expanded eligibility) and did not allow for the different life gained profiles of eligible individuals who did and did not attend screening.

Our study has many strengths. We performed the first analysis looking at the risk and benefit profiles of the individuals who are being screened, and we evaluated the lost opportunity from the limited uptake to date. We evaluated lung cancer screening uptake using data from a nationally representative survey, overall and by a range of characteristics. Self-reported smoking data have been shown to be reliable (25). The limitations of the self-report of lung cancer screening attendance in BRFSS have been discussed extensively (6,8,9,23), with the wording in 2022 still not distinguishing between individuals who had a chest CT specifically for lung cancer screening and those who have a chest CT for diagnostic purposes; our screening uptake results are therefore likely an overestimate. Individuals may have misremembered what test they had and reported a computed axial tomography scan or low-dose CT when they received a different test. BRFSS is a telephone survey with a median response rate for all states, territories, and Washington, DC, of 45.1% in 2022 (range = 22.8%-66.8%) (11), which may indicate considerable volunteer bias, though the proportion of individuals aged 50-79 years who self-reported ever-smoking (43.9%) was consistent with those reported in other national surveys (NHIS 2022: 41.8%; National Health and Nutrition Examination Survey 2017-2020: 46.2%). Not all variables required to estimate the risk and benefit of an individual were included in BRFSS, so some imputation was required to account for missing data. Because age was only available in 5-year age groups, we were not able to include individuals aged 80 years in our analysis, though they may be eligible for lung cancer screening. Our estimates of lung cancer deaths prevented and life-years gained from attending screening assume individuals attend 3 annual screens, as in the NLST.

This nationally representative study shows that among individuals eligible for lung cancer screening under USPSTF-2021, those who attended screening in the past year had a higher risk of lung cancer death and a higher benefit from attending screening, measured by life-years gained from attending screening compared with eligible individuals who were not screened. Although this is encouraging, only 18% of preventable lung cancer deaths are being prevented among the 10% of individuals aged 50-79 years who ever-smoked who attended screening in the past year, with the potential for 69% to be prevented if all 38% who are eligible under USPSTF-2021 attended.

Data availability

BRFSS data are available from https://www.cdc.gov/brfss/annual_data/annual_data.htm.

Author contributions

Elizabeth Y. Zhang, BA (Data curation; Formal analysis; Methodology; Writing—original draft),Li C Cheung, PhD (Methodology; Software; Writing—review & editing), Hormuzd A. Katki, PhD (Conceptualization; Methodology; Supervision; Writing—review & editing), Barry I. Graubard, PhD (Methodology; Writing—review & editing), Ahmedin Jemal, PhD (Conceptualization; Writing—review & editing), Anil K Chaturvedi, PhD (Conceptualization; Writing—review & editing), and Rebecca Landy, PhD (Conceptualization; Data curation;

Formal analysis; Methodology; Supervision; Writing—original draft).

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Conflicts of interest

The Lung Cancer Death Risk Assessment Tool (LCDRAT) and Life-Years From Screening–CT (LYFS-CT) were previously proposed by co-authors of this manuscript.

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